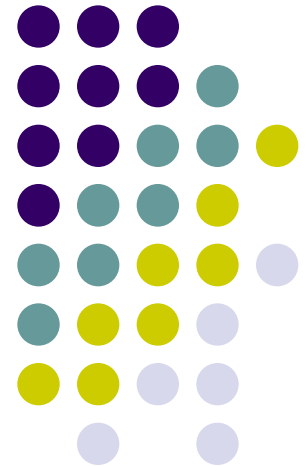


End-use Energy Demand Analysis

INTRODUCTION

Energy Modeling Workshop/Training
Bucharest, Romania
May 19-20, 2014

Sunil Malla
Consultant, The World Bank



Outline



I. Introduction

(Energy system, Energy statistics/indicators)

II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

V. End-use energy demand (model and examples)

Outline



I. Introduction

(Energy system, Energy statistics/indicators)

II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

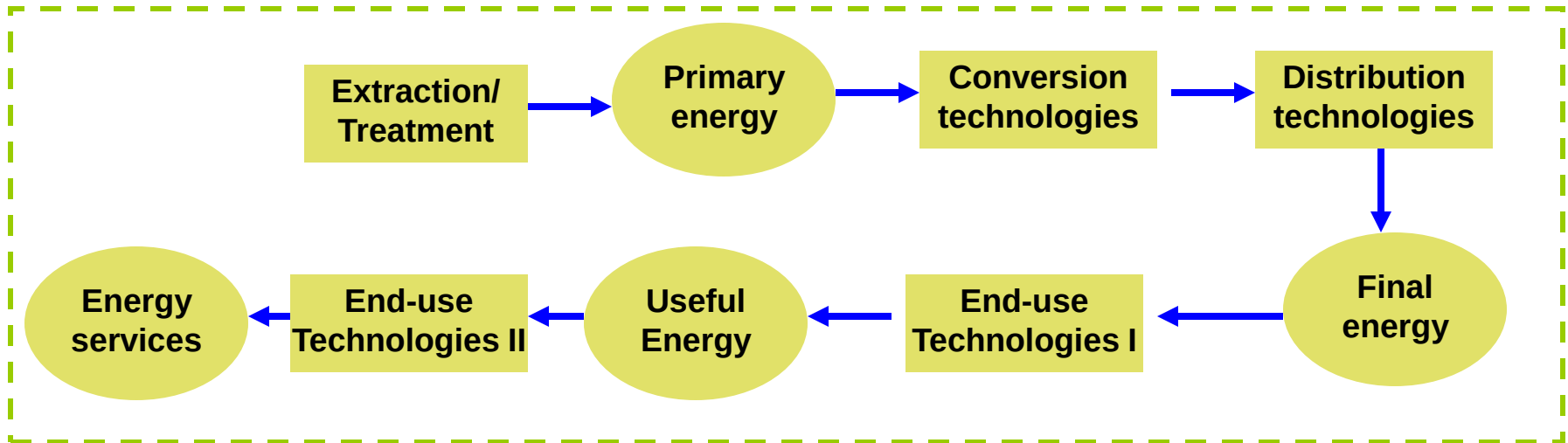
V. End-use energy demand (model and examples)

What is an energy system?

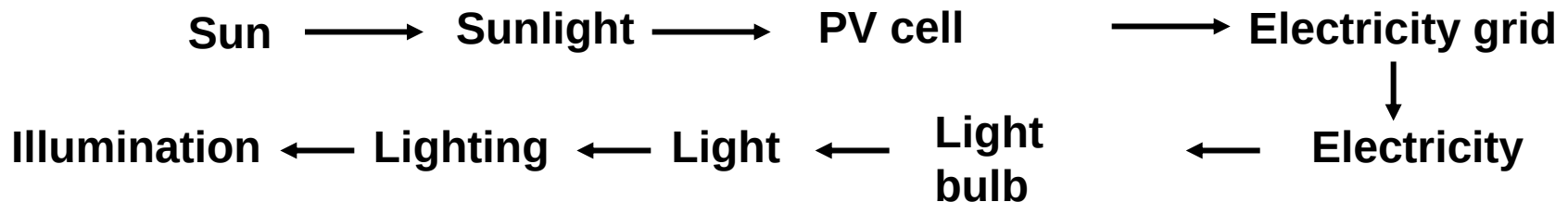


- ❖ An **energy system** comprises an energy supply and end-use sector.
- ❖ The **supply sector** consists of processes for extracting, converting and delivering energy resources into more desirable and sustainable forms to end users.
- ❖ The **end-use sector** provides services such as cooking, lighting, heating, cooling and mobility.
- ❖ The purpose of the energy system is to fulfill the demand for **energy services**.

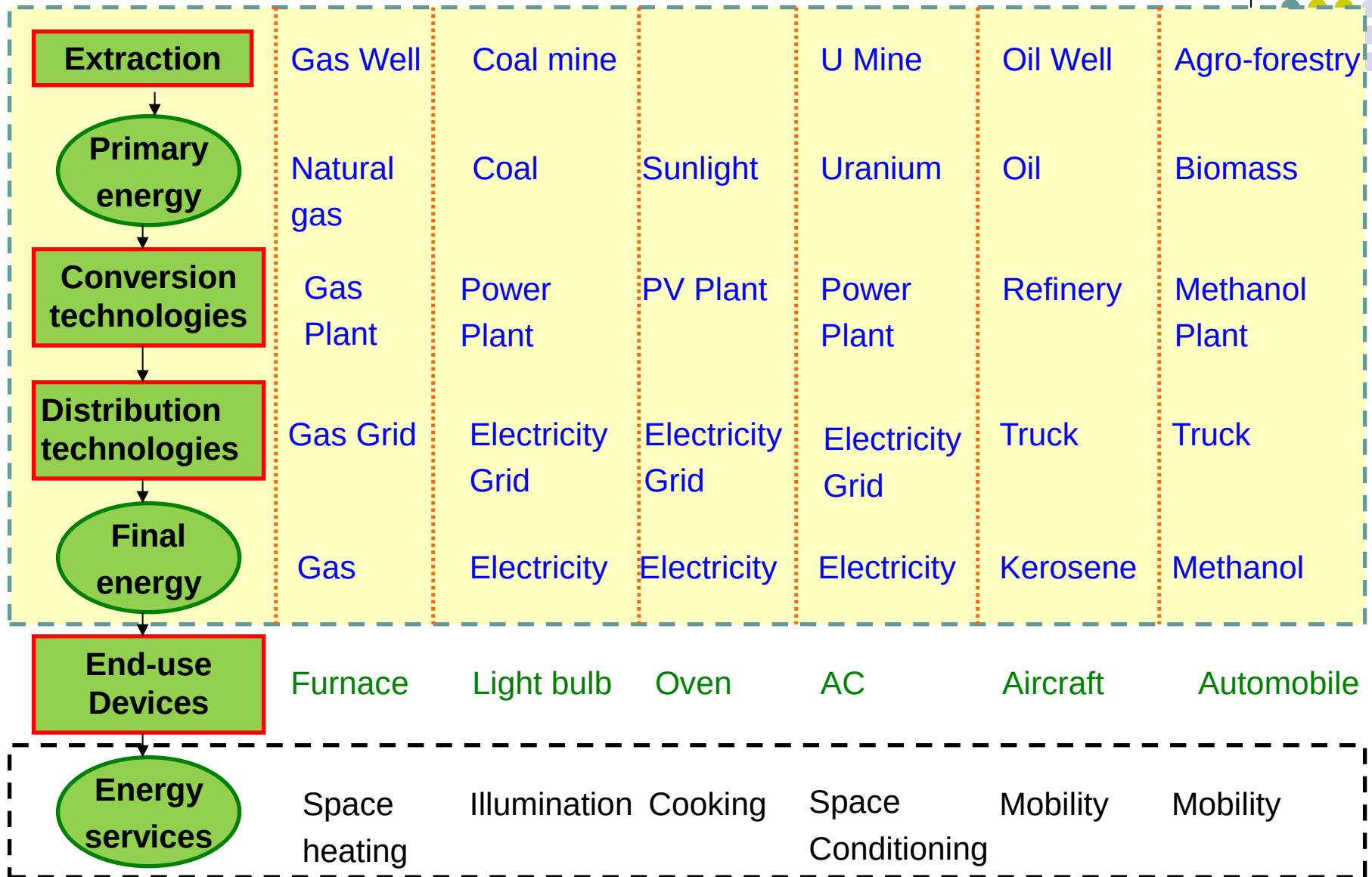
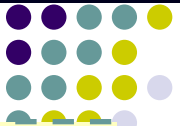
The structure of energy system, defined by the parts of the energy system and their interactions, is best represented by a so-called *reference energy system* (RES).



Example: 



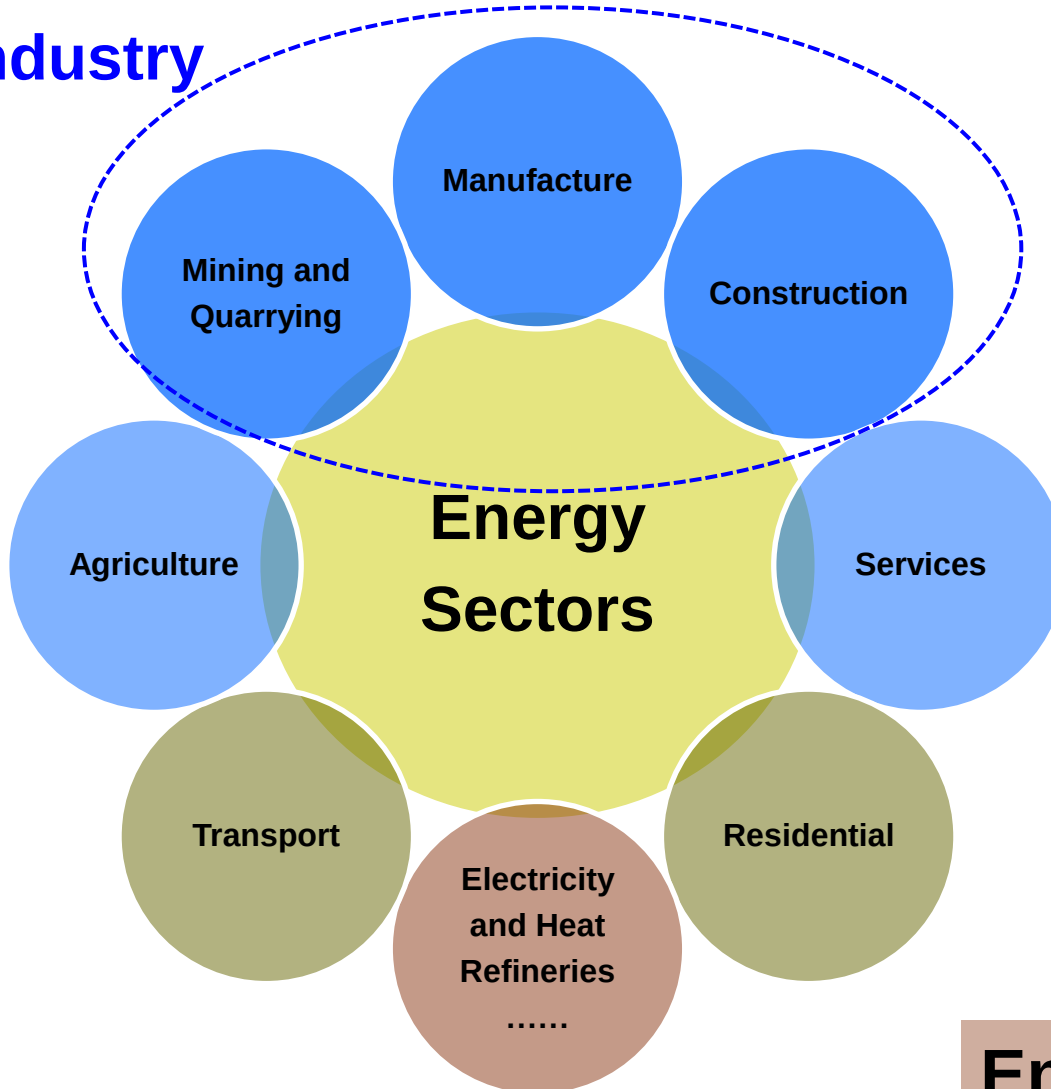
Reference Energy System (RES)



Energy Sector and the Economy



Industry



Economic sectors



Energy demand
sectors



Non-economic sectors

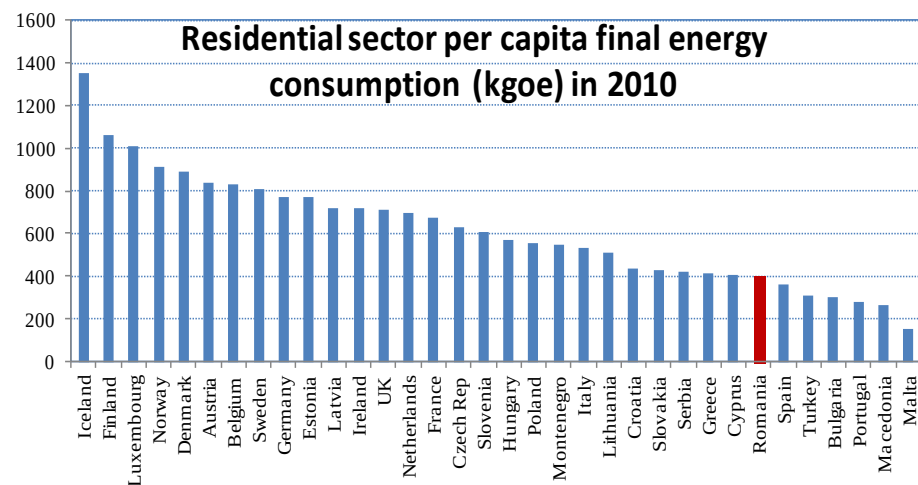
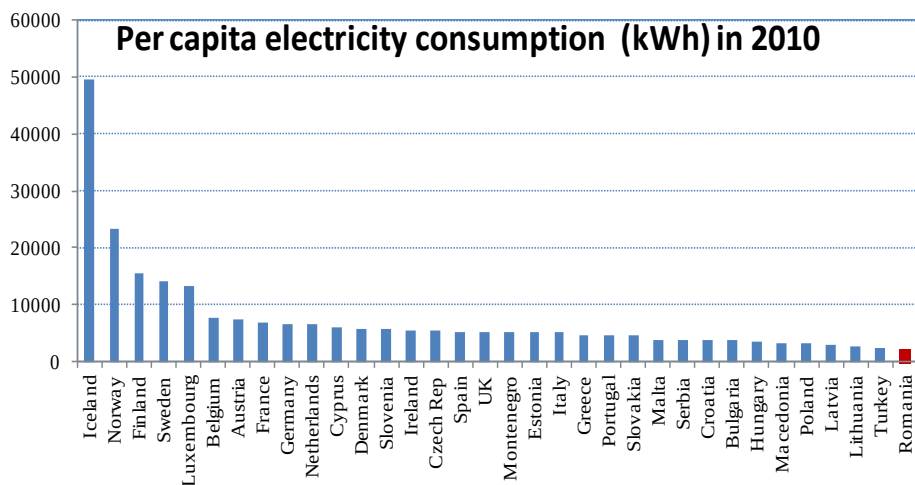
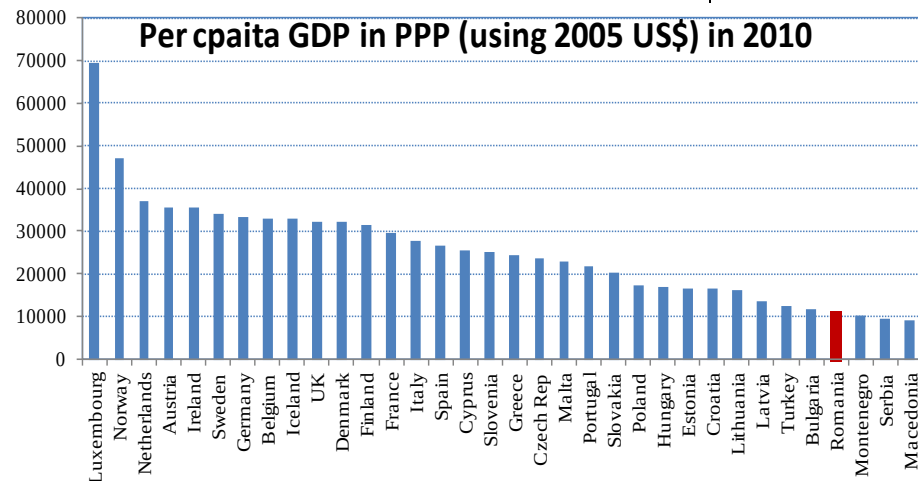
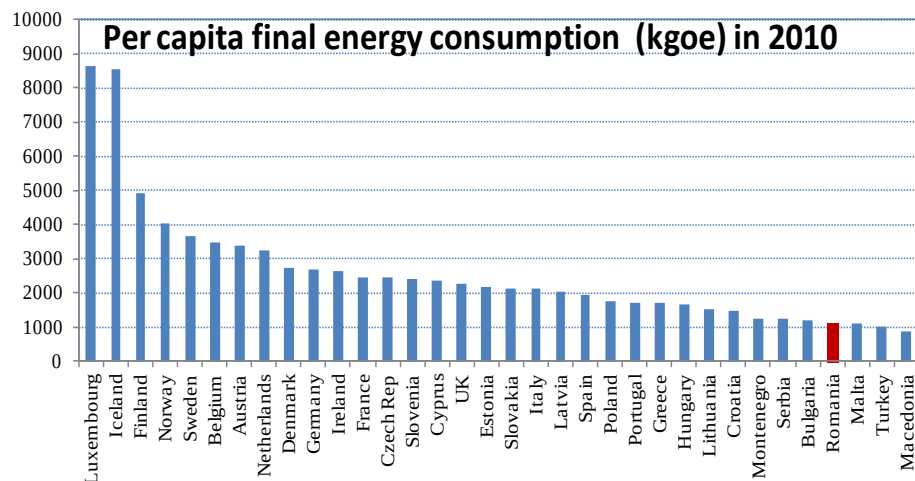
Energy supply sector

Importance of energy statistics



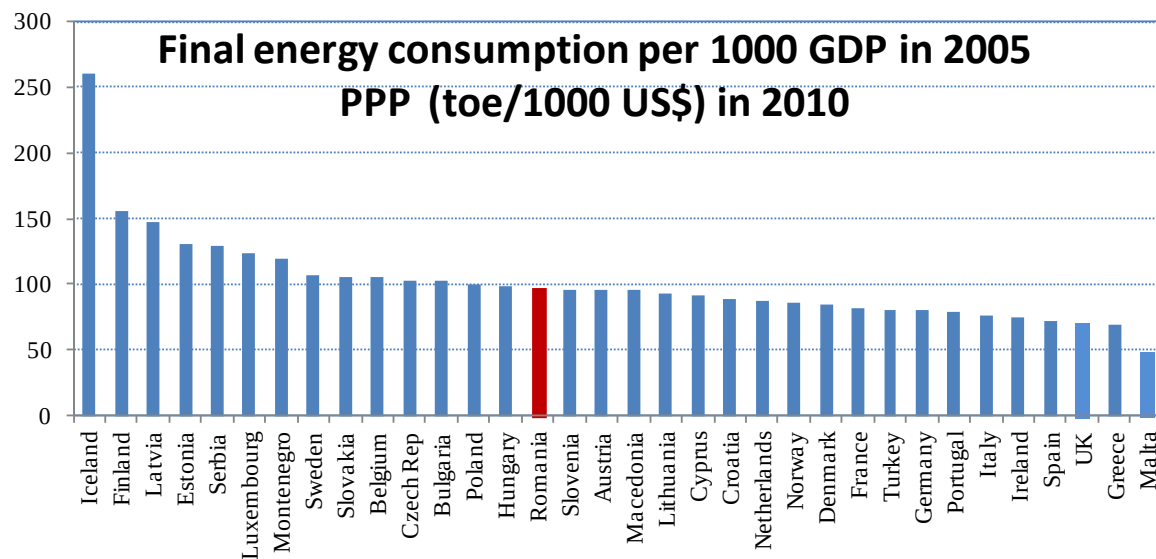
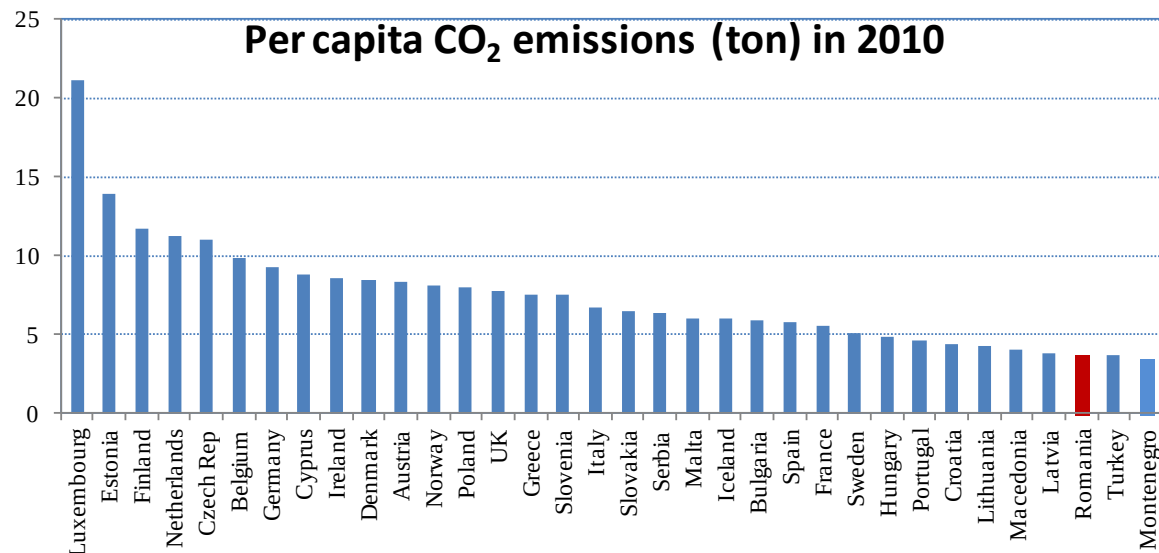
- ❖ Energy is an essential component of country's economy
- ❖ Access to energy
- ❖ Energy security (energy dependency)
- ❖ Fuel poverty
- ❖ Environment
- ❖ Energy efficiency
- ❖ Low carbon economy
- ❖ International/regional/national obligations and commitments
 - ❑ Obligation to have a minimum share of electricity consumption from renewable energy sources (**regional**)
 - ❑ UNFCCC national communication (**international**)
 - ❑ National renewable energy policy (**national**)

Selected energy indicators in EU



Source: Eurostat, World Bank, IEA.

Selected energy indicators in EU



Source: Eurostat, IEA.

Policy and measures (Romania)



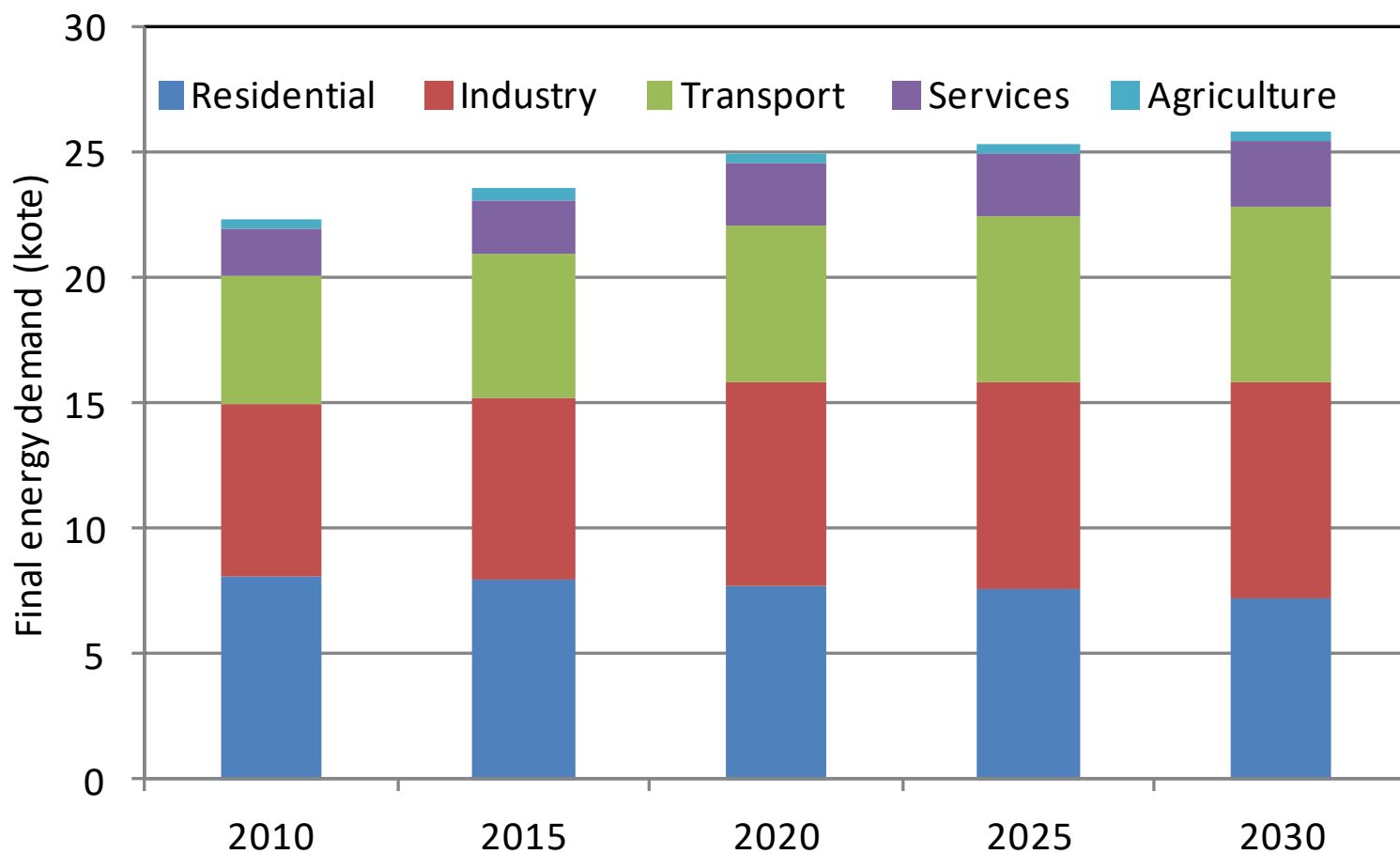
EU Directive	Policy/Measure
2006/32/EC	Energy end-use efficiency and energy services
2005/32/EC	Establish framework for the setting of eco-design requirements for energy-using products
2009/28/EC	promotion of the use of energy from renewable sources
2004/8/EC	promotion of cogeneration based on a useful heat demand in the internal energy market
2009/33/EC	promotion of clean and energy-efficient road transport vehicles
2009/443/EC	establishing the emission performance standards for new passenger cars
2010/40/EU	framework for the deployment of ITS in the field of road transport and for interfaces with other modes of transport

24% of the final energy consumption will be cover by renewable energy resources in 2020. (**National action plan for renewable energy**)

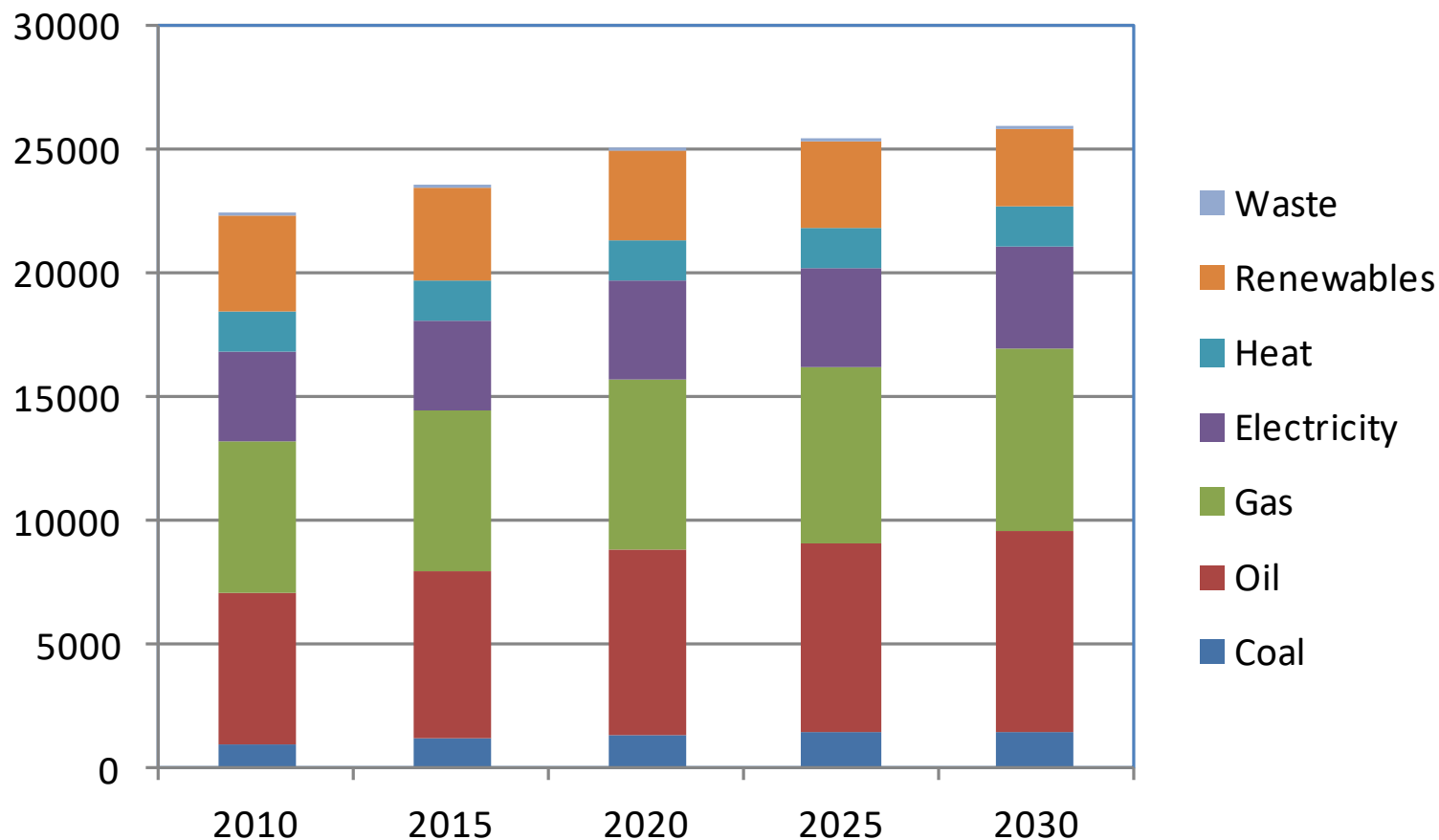
The **2nd National Action Plan on Energy Efficiency for 2011 – 2020** consider promotes measures for efficient use of energy (**20% reduction in year 2020**)

Implementation of **biofuels Directive** (Transport)

Projections of final energy demand by sector in Romania



Projections of final energy demand by energy type in Romania



Outline



I. Introduction

(Energy system, Energy statistics/indicators)

II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

V. End-use energy demand (model and examples)

Energy balance: source of energy information

What are the available energy sources?

Supply

Transformation and energy industries own use

Non-energy use

What is the total energy required? (Demand)

End-use energy sectors

Statistical differences

ROMANIA

Year : 2010

	Total all products	Hard coal	Coke	Lignite	Peat	Brown coal briquettes
	1000 toe					
Primary production	27,705	2		5,901	1	
Recovered products	67	0	0	0	0	0
Imports	11,711	519	736	21	8	0
Stock change	163	-6	-30	-74	0	0
Exports	3,975	46	3	0	0	0
Bunkers	16	0	0	0	0	0
Direct use	0					
Gross inland consumption	35,655	469	683	5,848	8	0
Transformation input	23,689	225	223	5,619	0	0
Main act. thermal power stations	7,782	183	0	5,227	0	0
Autoprod. thermal power stations	645	39	0	388	0	0
Nuclear power stations	2,968					
Briquetting plants	0	0	0	0	0	0
Coke-oven plants	2	2	0	0	0	0
Blast-furnace plants	223	0	223	0	0	0
Gas works	0	0	0	0	0	0
Refineries	11,350					
District heating plants	689	0	0	3	0	0
Transformation output	17,394		0			0
Main act. thermal power stations	4,071					
Autoprod. thermal power stations	286					
Nuclear power stations	999					
Briquetting plants	0					
Coke-oven plants	0		0			
Blast-furnace plants	223					
Gas works	0		0			
Refineries	11,327					
District heating plants	487					
Exchanges and transfers, returns	358					
Interproduct transfers	0					
Products transferred	0					
Returns from petrochem. industry	358					
Consumption of the energy branch	3,291	11	0	79	0	0
Distribution losses	1,425	5	1	2	0	0
Available for final consumption	25,002	229	458	147	8	0
Final non-energy consumption	2,308	0	0	0	0	0
Industry	1,681					
of which: chemical/petrochemical	1,064	0	0	0	0	0
Final energy consumption	22,509	180	599	152	8	0
Industry	6,909	180	599	142	0	0
Iron & steel industry	2,023	138	598	0	0	0
Non-ferrous metal industry	0	0	0	0	0	0
Chemical industry	2,056	37	0	90	0	0
Glass, pottery & building mat. industry	543	4	1	51	0	0
Ore-extraction industry	48	0	0	0	0	0
Food, drink & tobacco industry	557	0	1	0	0	0
Textile, leather & clothing industry	181	0	0	0	0	0
Paper and printing	165	0	0	0	0	0
Transport equipment	114	0	0	0	0	0
Machinery	365	0	0	0	0	0
Wood and wood products	263	0	0	0	0	0
Construction	408	0	0	0	0	0
Non specified (Other)						
Transport	5,013	0	0	0	0	0
Railways	221	0	0	0	0	0
Road transport	4,404					
International aviation	166					
Domestic aviation	111					
Domestic navigation	59	0	0	0	0	0
Other sectors	10,588	0	0	10	8	0
Households	8,102	0	0	10	0	0
Services	1,881	0	0	1	0	0
Agriculture/Forestry	391	0	0	0	8	0
Fishing	0	0	0	0	0	0
Statistical difference	186	50	-141	-5	0	0

Source: EUROSTAT.

Total final energy consumption =



❖ Non-energy use +

Eurostat / IEA

❖ Final energy consumption (FEC) +

❑ Industry (13 branches)

❑ Transport (6 modes)

❑ Households

❑ Services

❑ Agriculture

❑ Fishing

❖ Iron & steel industry

❖ Non-ferrous metal industry

❖ Chemical industry

❖ Glass, pottery & building mat. industry

❖ Ore-extraction industry

❖ Food, drink & tobacco industry

❖ Textile, leather & clothing industry

❖ Paper and printing

❖ Transport equipment

❖ Machinery

❖ Wood and wood products

❖ Construction

❖ Non specified (Other)

❖ Railways

❖ Road transport

❖ International aviation

❖ Domestic aviation

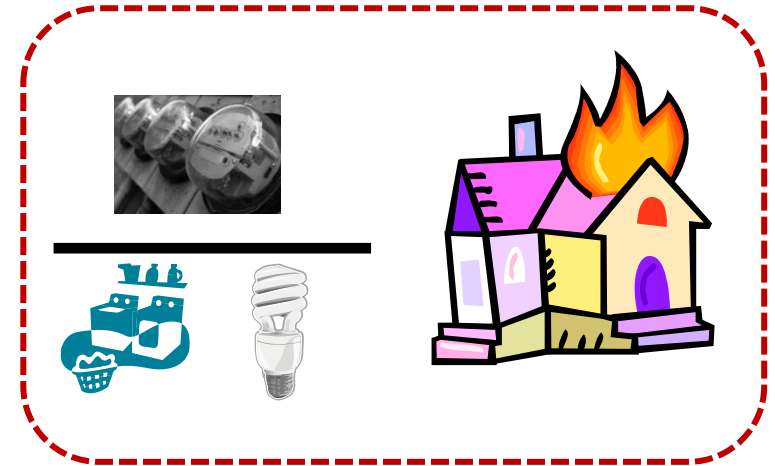
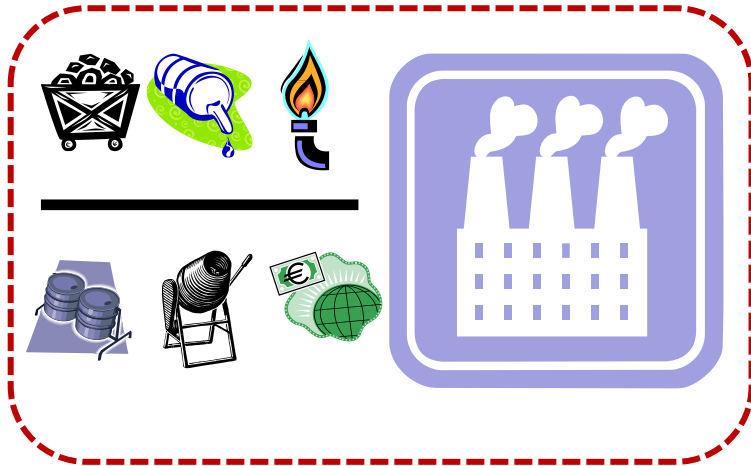
❖ Domestic navigation

❖ Pipeline transport

❖ Statistical differences +

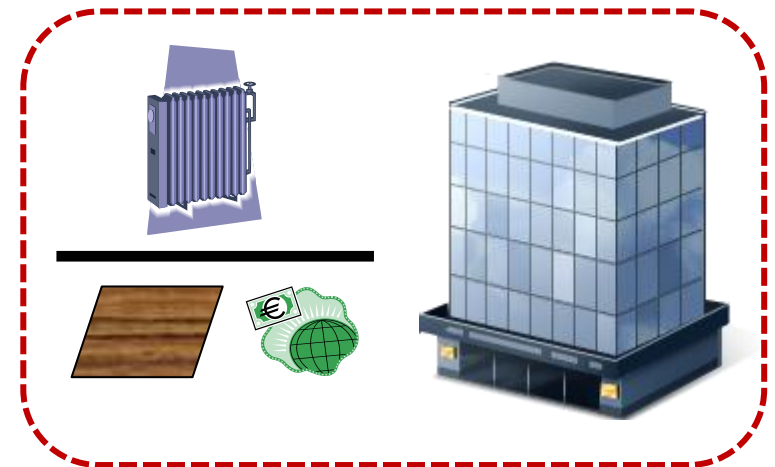
No breakdown available

Non only energy but activity data is also necessary

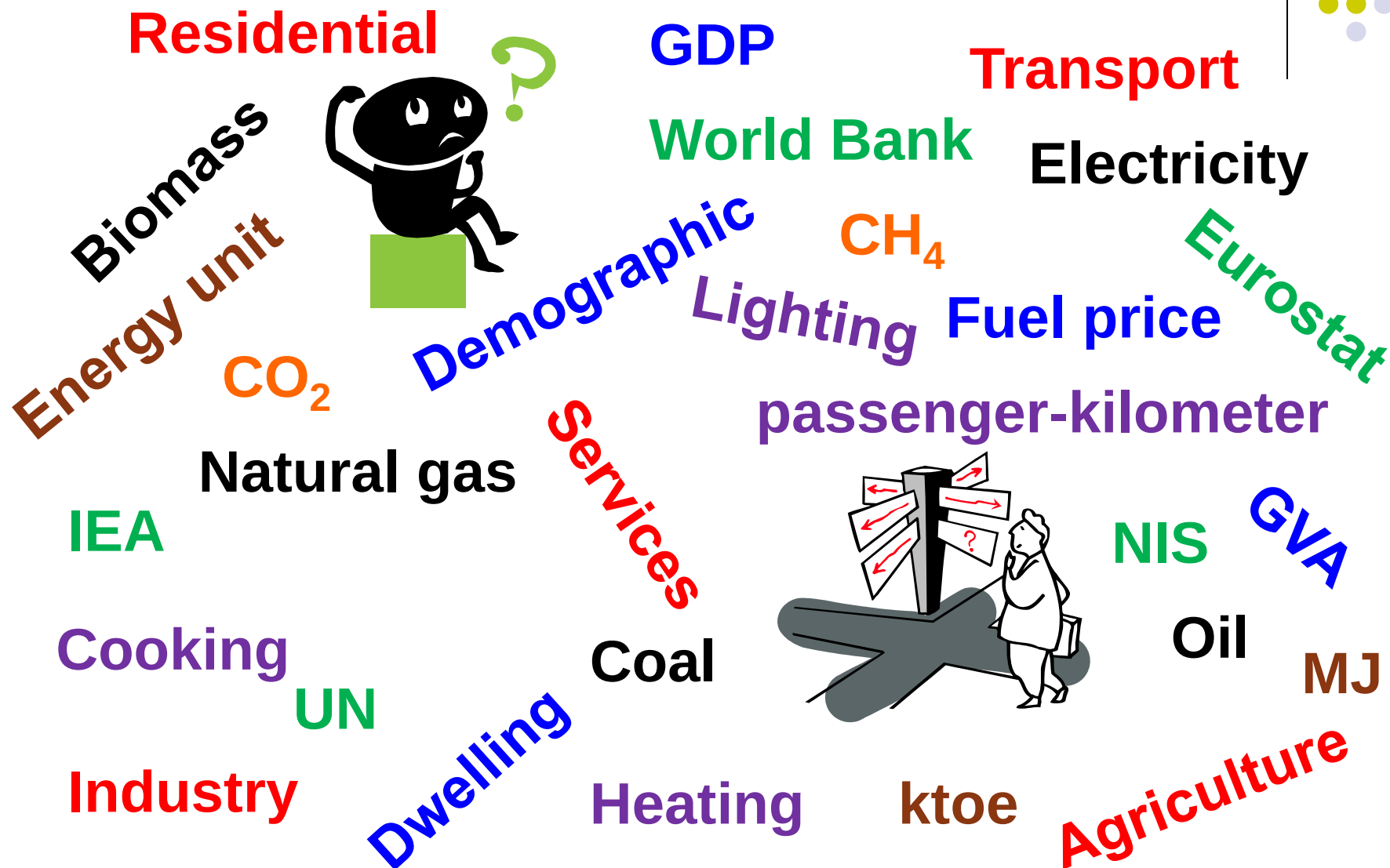


Energy

Activity



What data to collect and how?



Approach for data collection



- ❖ **Understand** national energy situation
- ❖ **Set** priorities (Residential? Industry? Base year, Time horizon.....)
- ❖ **Identify** targets and objectives (national/regional plans)
- ❖ **Consider** cost (budget)
- ❖ **Set** time frame
- ❖ **Be** practical collecting on necessary data (surveys, published sources, modeling, metering....)
- ❖ **Limit** to reliable and official data sources
- ❖ **Decide** methodology

Then proceed

End-use Energy Demand Analysis



	MACRO	Residential	Services	Transport	Industry	Agriculture
Sector/Sub-sector/ End-use	Residential Services Transport Industry Agriculture	Cooking Lighting Space heating Cooling Water heating Appliances - Refrigerator - Washing m/c - Dish washer - TVs - Microwave - Others	Lighting Space heating Cooling Water heating Appliances	Passenger - Road (car/bus/minibus) - Railways - Air - Waterways Freight - Road (Lorries/tractors) - Railways - Air - Waterways - Maritime - Pipeline	Iron and steel Non-ferrous metal Chemical & petro-chemical Non-metallic minerals Food & tobacco Textile & leather Pulp & paper Transport equipment Machinery Wood & wood products Mining and quarrying Construction Others	Agriculture /Forestry Fisheries
Technical/socio-economic data	Final energy consumption GDP GVA Population	EC by end-use Dwelling stock Floor area Appliances stock Equipment rate Degree days	EC by end-use GVA Dwelling stock Floor area Employment	EC by fuel/mode type Stock of motor vehicle Registration of vehicle vkm, pkm, tkm Fuel economy No. of passenger	EC by branch GVA Production by branch	EC by branch GVA
Data source	NIS, Eurostat, WDI, IEA	NIS, Eurostat, IEA, HH survey, Assumption	NIS, Eurostat, IEA, Assumption	NIS, Eurostat, IEA, Assumption	NIS, Eurostat, IEA, Assumption	NIS, Eurostat, Assumption

Outline



I. Introduction

(Energy system, Energy statistics/indicators)

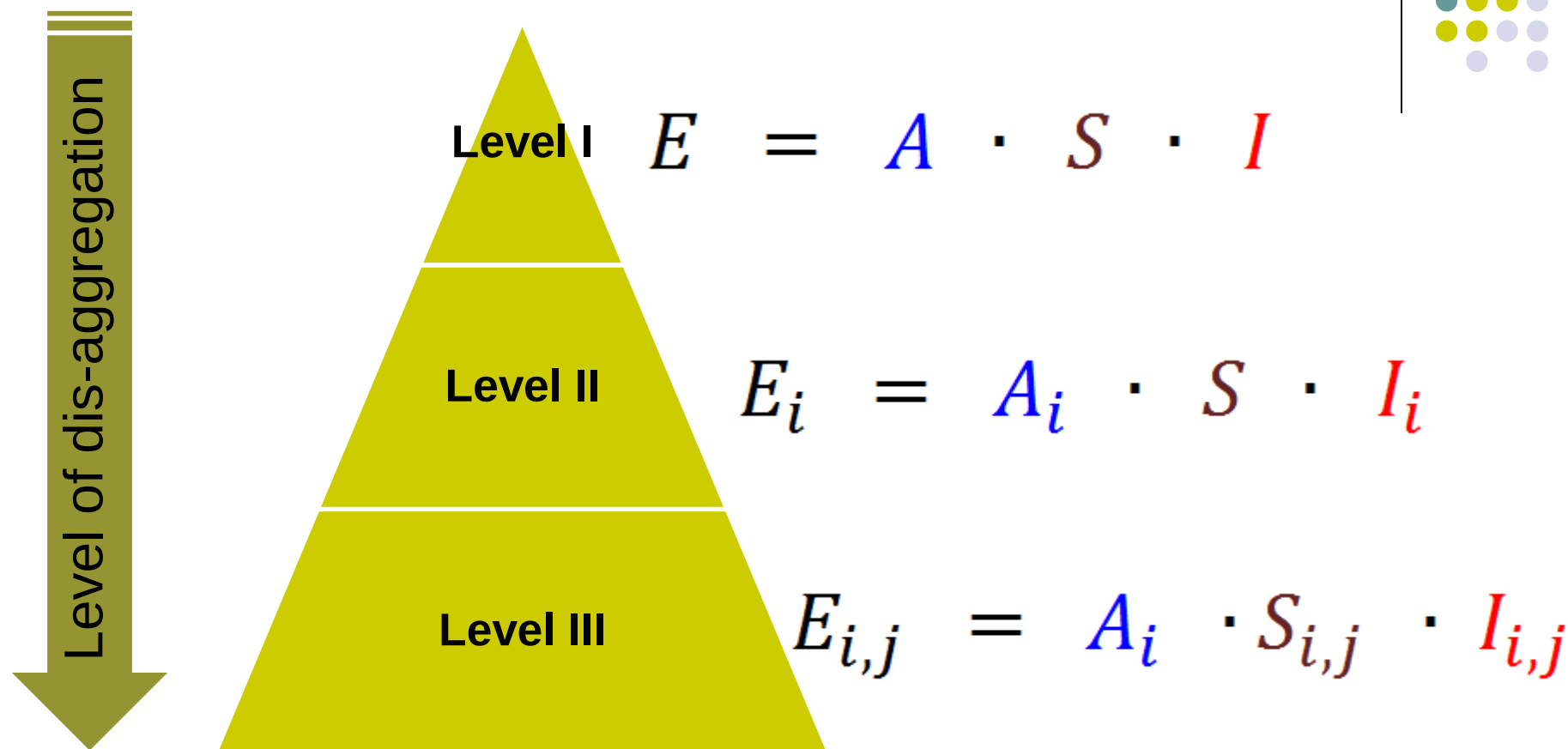
II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

V. End-use energy demand (model and examples)

Methodology



A = Activity

i = sector

S = Structure

j = sub-sector or end-use

I = Energy intensity

Example



Sector	Sub-sector	A	S	I or (UEC)
Residential				
	Space heating Water heating	Floor area Population	Floor area/dw Population/dw	kgoe/floor area kgoe/person
Transport				
	Passenger cars Freight lorries	passenger-km ton-km	share of pkm share of tkm	kgoe/pkm kgoe/tkm
Industry				
	Iron and steel Pulp and paper	Value added/ Production Value added	share of VA share of VA	kgoe/VA kgoe/ton of steel Kgoe/VA

Outline



I. Introduction

(Energy system, Energy statistics/indicators)

II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

V. End-use energy demand (model and examples)

Useful resources



1. National Institute of Statistics (NIS)

- ❑ TEMPO-Online time series [<https://statistici.insse.ro/shop/?lang=en>]
- ❑ Statistical yearbook [<http://www.insse.ro/cms/en/content/statistical-yearbook-2012>]

2. Ministry of Environment and Climate Change, Romania

- ❑ UNFCCC National Communication several editions [Latest 6th NC; December 2013]

3. Eurostat

- ❑ Energy database
[<http://epp.eurostat.ec.europa.eu/portal/page/portal/energy/data/database>]
- ❑ Transport database
[<http://epp.eurostat.ec.europa.eu/portal/page/portal/transport/data/database>]

4. The World Bank

- ❑ EFFECT model [<http://esmap.org/EFFECT>]
- ❑ WDI Databank [<http://data.worldbank.org/data-catalog/world-development-indicators>]

5. International Energy Agency (IEA) [<http://www.iea.org>]

- ❑ Energy Statistics Manual
- ❑ Energy balance of OECD countries; CO₂ Emissions from Fuel Combustion

6. ENPEP/MAED [IAEA] [http://www-pub.iaea.org/MTCD/Publications/PDF/te_963_prn.pdf]

7. ODYSSEE-MURE [ENERDATA/ISIS] [<http://www.odyssee-mure.eu/>]

Outline



I. Introduction

(Energy system, Energy statistics/indicators)

II. Type and approach collecting energy data

III. Methodology (Decomposition method)

IV. Data and data sources

V. End-use energy demand (model and examples)

EEDA Model demonstration



End-use Energy Demand Analysis

Romania

SECTOR

RESIDENTIAL
TRANSPORT

INDUSTRY
SERVICES
AGRICULTURE

TABLE OF CONTENT

Classification
Dataset
Assumption
Illustration

Energy and activity data for
passenger and freight transport



Transport
Socio economic and activity data
Energy demand forecast



Energy units: orders of magnitude



Symbol	Name	Magnitude	Number	Expression
P	Peta	10^{15}	1 000 000 000 000 000	quadrillion
T	Tera	10^{12}	1 000 000 000 000	trillion
G	Giga	10^9	1 000 000 000	billion
M	Mega	10^6	1 000 000	million
k	Kilo	10^3	1 000	thousand
h	Hecto	10^2	100	hundred
da	Deka	10^1	10	ten
-		1	1	one
da	Deci	10^{-1}	0.1	tenth
c	Centi	10^{-2}	0.01	hundredth
m	Milli	10^{-3}	0.001	thousandth
μ	Micro	10^{-6}	0.000 001	millionth
n	Nano	10^{-9}	0.000 000 001	billionth
p	Piko	10^{-12}	0.000 000 000 001	trillionth

Energy: conversion equivalent



To:	TJ	Gcal	Mtoe	MBtu	GWh
From:	multiply by:				
<i>Terajoule (TJ)</i>	1	238.8	2.388×10^{-5}	947.8	0.2778
<i>Gigacalorie</i>	4.1868×10^{-3}	1	10^{-7}	3.968	1.163×10^{-3}
<i>Mtoe*</i>	4.1868×10^4	10^7	1	3.968×10^7	11630
<i>Million Btu</i>	1.0551×10^{-3}	0.252	2.52×10^{-8}	1	2.931×10^{-4}
<i>Gigawatt-hour</i>	3.6	860	8.6×10^{-5}	3412	1

1 ktoe = **0.041868 PJ**

Net Calorific Value (NCV)

Gasoline: **44.75 GJ/ton**

Diesel: **43.38 GJ/ton**

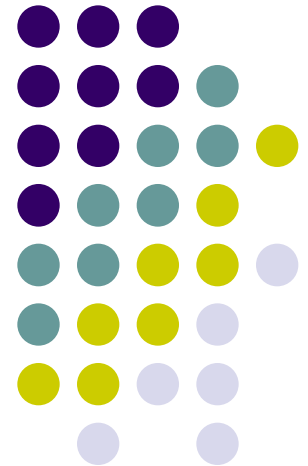
Net Calorific Value (NCV)

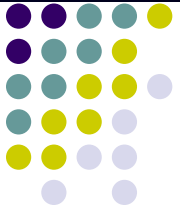
Gasoline: **1350 l/ton**

Diesel: **1185 l/ton**

Thank you !!

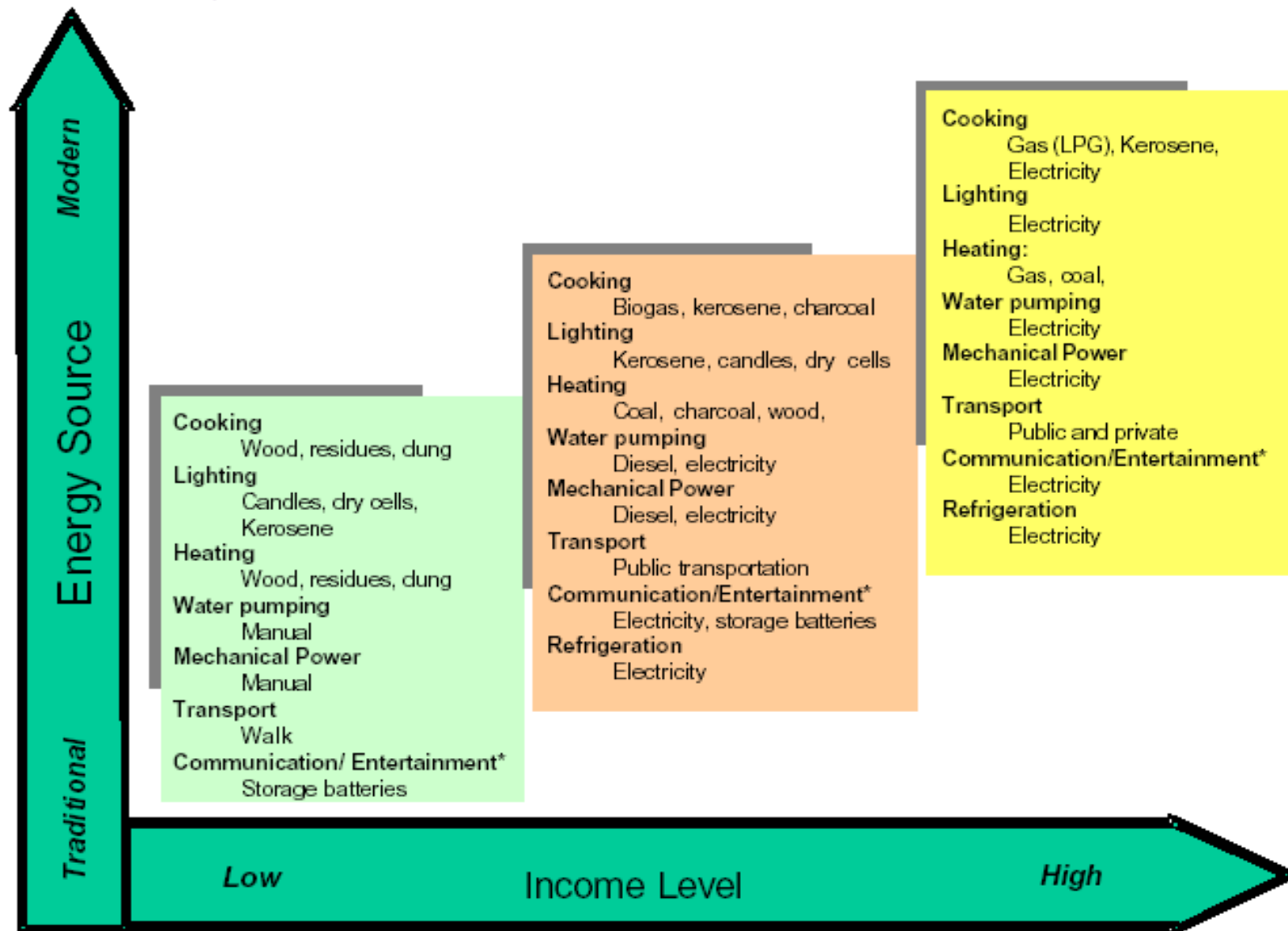
Questions/Comments





Extra slides

Energy ladder hypothesis



Key terms

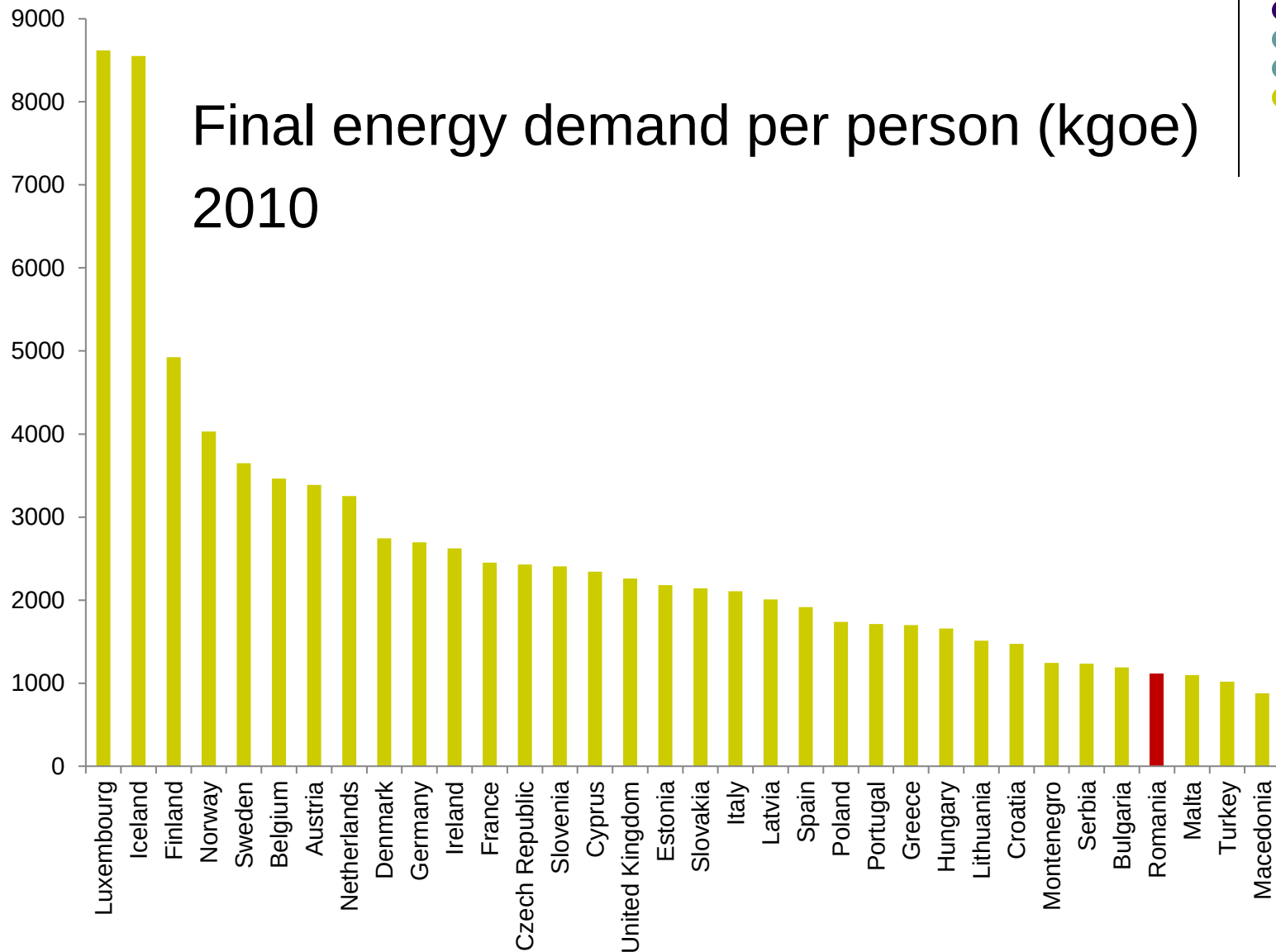


- ❖ **Useful Energy:** Delivered energy minus losses estimated for end use devices (water heater, cooking stoves, electrical appliances etc.).
- ❖ **Activity or Output:** Basic unit of accounting for which energy is used. For example, Floor space, value added, ton of steel production, population, passenger and freight kilometer travelled etc.
- ❖ **Energy Intensity:** Energy ‘consumed’ per unit of activity or output.
- ❖ **Structure:** Refers to the activity mix; e.g., modal mix (trucks, rail, ships) in travel, energy end uses in households and the shares of each sub-sector in total manufacturing value added.
- ❖ **Energy Services:** Implies actual services for which energy is used. Heating, cooking, lighting mobility etc. It is obtained from combined activity and structure measures.

Commonly used terms



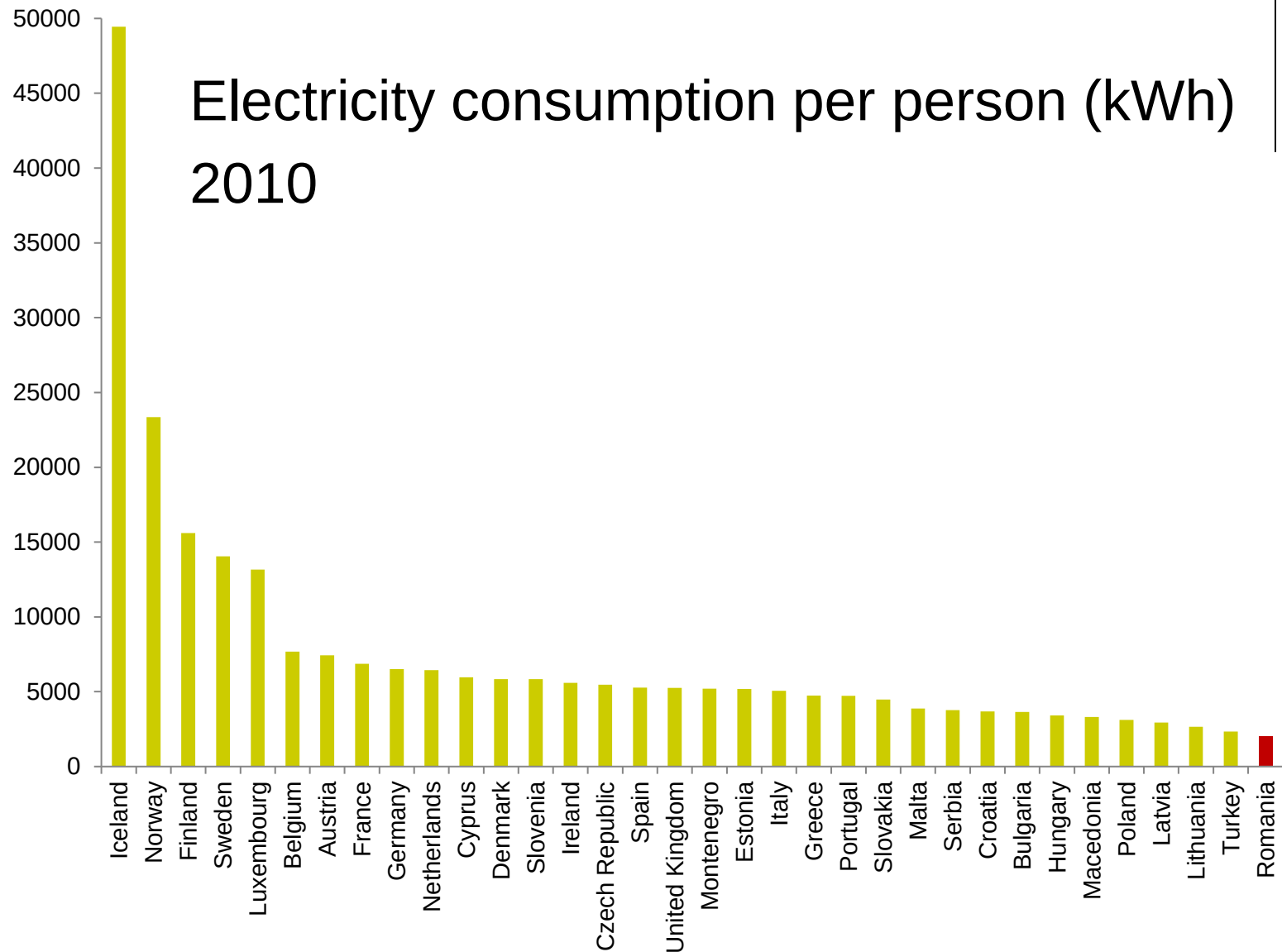
Term	Definition
pkm	Passenger-kilometer (pkm) is a unit of measure of passenger transport. 1 pkm represents the transport of 1 passenger over one kilometer.
tkm	Ton-kilometer (tkm) is an unit of measure of freight transport. 1 tkm represents the transport of 1 ton over one kilometer.
vkm/vkt	Vehicle kilometer travelled (vkt) is a unit of measurement representing the movement of a vehicle over 1 kilometer. Vehicle in this definition also includes ships, planes and railways.
l/100km	Fuel efficiency of motor vehicles
EI	Energy intensity
HDD	Heating degree days



Source: EUROSTAT.



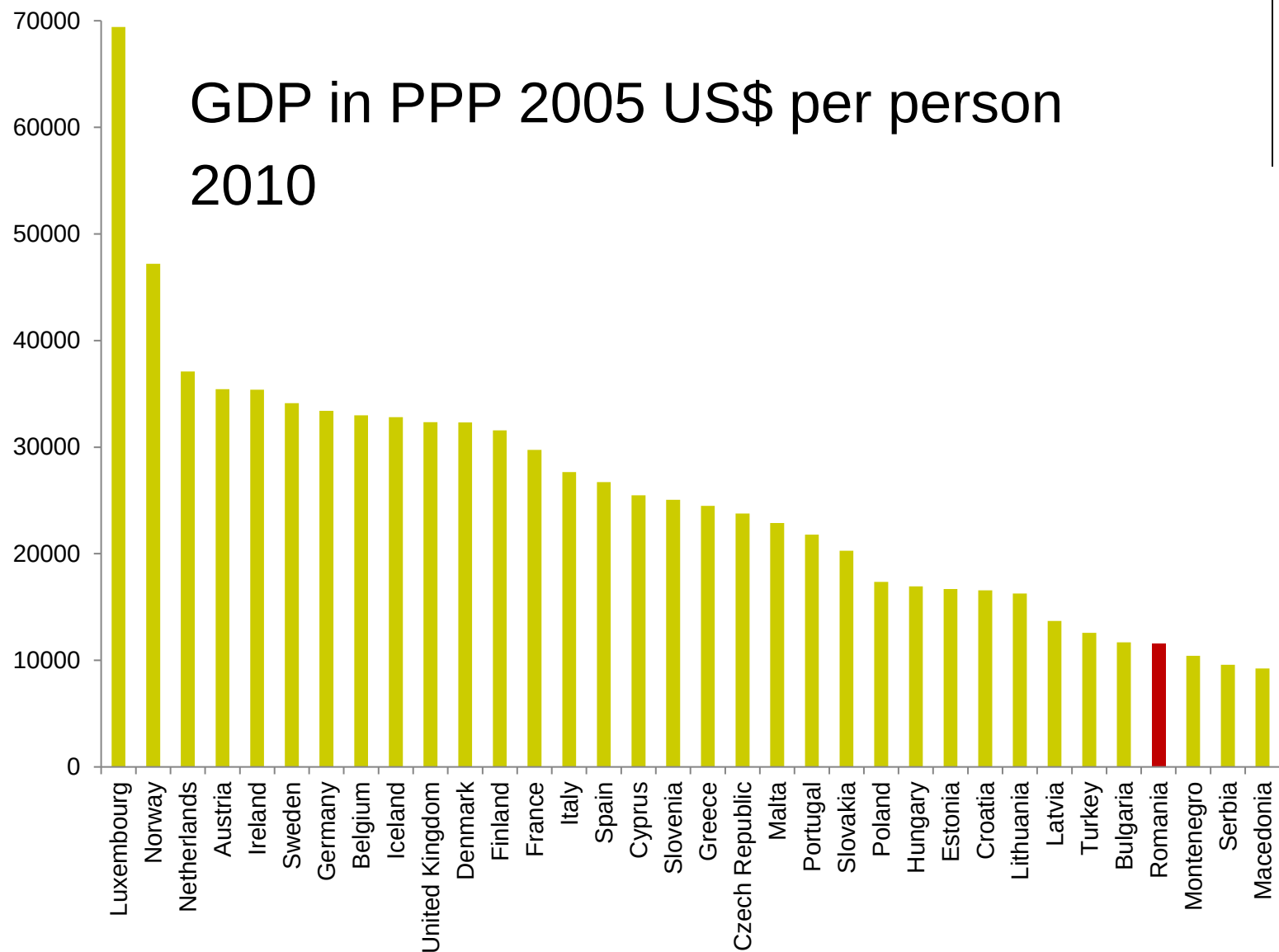
Electricity consumption per person (kWh) 2010



Source: EUROSTAT.



GDP in PPP 2005 US\$ per person 2010



Sources: EUROSTAT., IEA

